

PRAYAS PATNAIK

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| <https://prayaspatnaik21.github.io/>

EDUCATION

The University of Texas at Dallas - Richardson, TX
Master of Science in Computer Engineering

Dec 2023
GPA: 3.83/4

PROFESSIONAL EXPERIENCE

RFA Engineering

Software Engineer – Camera Imaging Team

Urbandale, Iowa
March 2025 - Present

- Working on **Image Quality** Optimization and **ISP Tuning** for image sensors.
- Designed PWL and **Auto Exposure Diagram** for the Sony Camera.
- Optimized **Reinhard Tone Mapping** in the Camera ISP pipeline by LUT Decimation.
- Automated perception quality metrics (**SNR, Delta-E**) calculations using eSFR Charts and ColorChecker boards for camera evaluation pipelines.
- Developed a GUI Based **Tuning application** to configure and validate sensor and ISP parameters.
- Integrated **Exposure Control Integration** for Onsemi image Sensor within the application pipeline.
- Developed and automated **Motion blur** analysis pipeline, computing Blur Edge Width using Siemens star charts across varying exposure times.
- Designed and optimized an automated **Color Correction Matrix (CCM)** by minimizing error between 24 patch Color Checker LAB values and reference standards.
- Developed an Integrating Sphere based **Lens Shading Correction (LSC)** characterization workflow, capturing uniform flat field images and analyzing center to edge luminance and color channel falloff to evaluate camera shading performance.
- Investigated **Ghosting Artifacts** in VD6763 based camera applications and designed characterization experiments to evaluate sensor/imaging parameters, establishing mitigation guidelines and recommended configurations for improve image quality.
- **Debugged and Fixed ISP pipeline issues** involving raw signal clipping, incomplete HDR short exposure fusion, and incorrect gain clipping in Reinhard Tone Mapping.
- Developed an automatic alpha estimation method for **Reinhard Tone Mapping** by extending the existing parametric estimation formulation.

Software Engineer (Automation Engineering Group) (Team – Furrow Vision)

May 2023 – March 2025

- Developed **Camera Switching Feature** for Gen5 Display on IPM (Image Processing Module) using **MQTT** for Furrow Vision Application.
- Designed and implemented **multi-threaded configuration interface** supporting multiple display generations.
- Implemented **Image Throttle Algorithm** for Application to operate on desired Frequency (Current 1Hz).
- **Integrated License Activation** Enablement Feature for the Application for subscription Policy. (**CAN protocol**).
- Built **MQTT based Communication service** to manage and process messages across multiple cameras and display clients.
- **Plugin Creation** to Calculate Average Automation Signal for Furrow Vision. (**RoboticsWorkbench**)
- Worked on Writing **Automation Test cases** for the Features Mentioned above for **Software Validation** using **Python**.
- Collaborated with the Verification Champion for the proper testing of the Features and **Field Testing**.
- **Feature Release Testing** and Resolve Defects.

John Deere India Pvt Ltd

Senior Engineer (Automation Engineering Group)

Pune, India

Jul 2019 - Dec 2021

- Optimized color conversion operations from **RGB/YCrCb** images to **NV12** images using **Neon Arm Compute Library** for the **Digital Streaming Service. (Arm Cortex A53) (Low level ML library)**
- Created Media Pipelines using Pipeline based multimedia framework **Gstreamer**.
- Implemented **OpenGL** Common Module Framework to stitch Left and Right Images from **Stereo Cameras** on **Arm Mali 400** GPU, resulted in CPU usage to **220% to 103%** and latency from **140ms to 95ms (32%)**. Developed Custom Shaders for Homography Operations.
- Worked on Python to write **automation test cases** for Software Validation and utilized **DISDATT tool** to facilitate testing.
- Employed **LTTNG** for **profiling code** and optimizing performance. Successfully identified bottlenecks in the code.
- Implemented **Bash scripting** to generate **code coverage** reports with **LLVM** and automate the task in **Jenkins**.